

ENGR 102 Summer Bridge

Day 3 Instructor Key

Raw Input, Type Evidence, Calculations, and F-Strings

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace raw text, conversion, and formatted output.

4 points

```
raw = "18.5"
value = float(raw)
adjusted = value + 1.5
print(type(raw).__name__, type(value).__name__)
print(f"{adjusted:.1f}")
```

Give the type and value at each named state, then write exact output.

Answer and explanation

raw remains the string 18.5. value is the float 18.5, and adjusted is the float 20.0. Exact output lines are str float and 20.0.

Question 2. Separate truthiness from an approval rule.

4 points

```
approved_text = "no"
print(bool(approved_text))
print(approved_text == "yes")
```

Give both results and explain why a nonempty input string is not reliable approval evidence.

Answer and explanation

A nonempty string is truthy, so `bool(approved_text)` is `True` even when the text is no. The explicit comparison to yes is `False`. Exact output: `True`, then `False`.

Question 3. Evaluate a multi-step Python math expression.

4 points

```
side_a = 5
side_b = 12
diagonal = (side_a ** 2 + side_b ** 2) ** 0.5
print(f"{diagonal:.2f}")
```

Show the squared terms, the sum, the square-root step, and exact output.

Answer and explanation

The squared terms are 25 and 144, whose sum is 169. Raising 169 to 0.5 gives 13.0. Two-decimal formatting prints 13.00.

Question 4. Program construction**8 points**

Write one complete Python program that satisfies every requirement.

- Read distance and time as floating-point inputs.
- If time is not positive, print invalid and do not calculate speed.
- Otherwise compute $\text{speed} = \text{distance} / \text{time}$.
- Print Speed: followed by the result to exactly two decimal places.

Answer and explanation

```
distance = float(input())
time = float(input())

if time <= 0:
    print("invalid")
else:
    speed = distance / time
    print(f"Speed: {speed:.2f}")
```

The time guard precedes division, so zero and negative denominators never reach the calculation.

For distance 125 and time 10, speed is 12.5 and the f-string prints Speed: 12.50. For time 0, the only output is invalid.

Scoring guidance

2 input/conversion, 2 protective time guard, 2 correct calculation/branch placement, 2 exact label and precision.